

Physical Chemistry (I)

Lecture 15



In the Last Class

- Phase boundaries
 - Clapeyron equation
 - Solid-liquid boundaries
 - Liquid-gas and solid-gas boundaries
- Simple mixtures: general aspects
 - Thermodynamics of a multi-component system
 - Partial molar quantities



Phase Boundaries

When two phases α and β are at equilibrium,

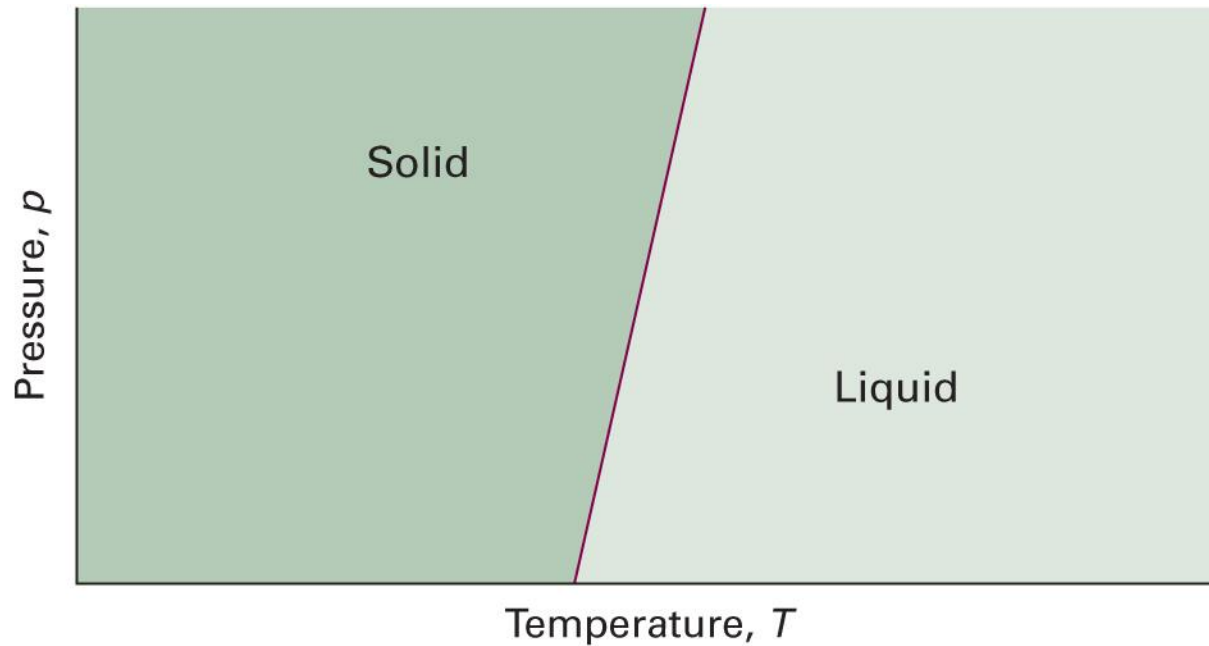
$$\mu(\alpha; p, T) = \mu(\beta; p, T)$$

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}, \quad \frac{dT}{dp} = \frac{\Delta_{\text{trs}}V}{\Delta_{\text{trs}}S}$$

(Clapeyron equation)

Solid-Liquid Boundary

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{T_1 \Delta_{\text{fus}}V} (T_2 - T_1)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.5)

Clausius-Clapeyron equation

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{RT^2}$$

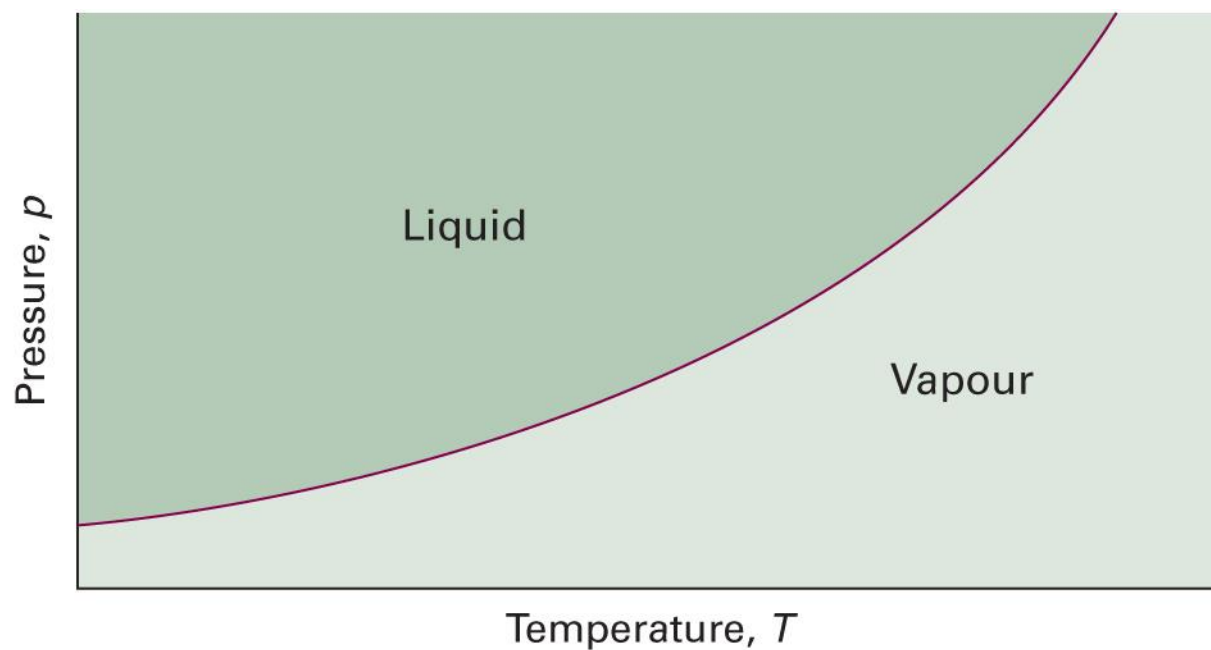
(Clausius-Clapeyron equation)

$$p_2 = p_1 e^{A-B/T_2}$$

where

$$A = \frac{\Delta_{\text{vap}}H}{RT_1}, \quad B = \frac{\Delta_{\text{vap}}H}{R}$$

Liquid-Gas Boundary

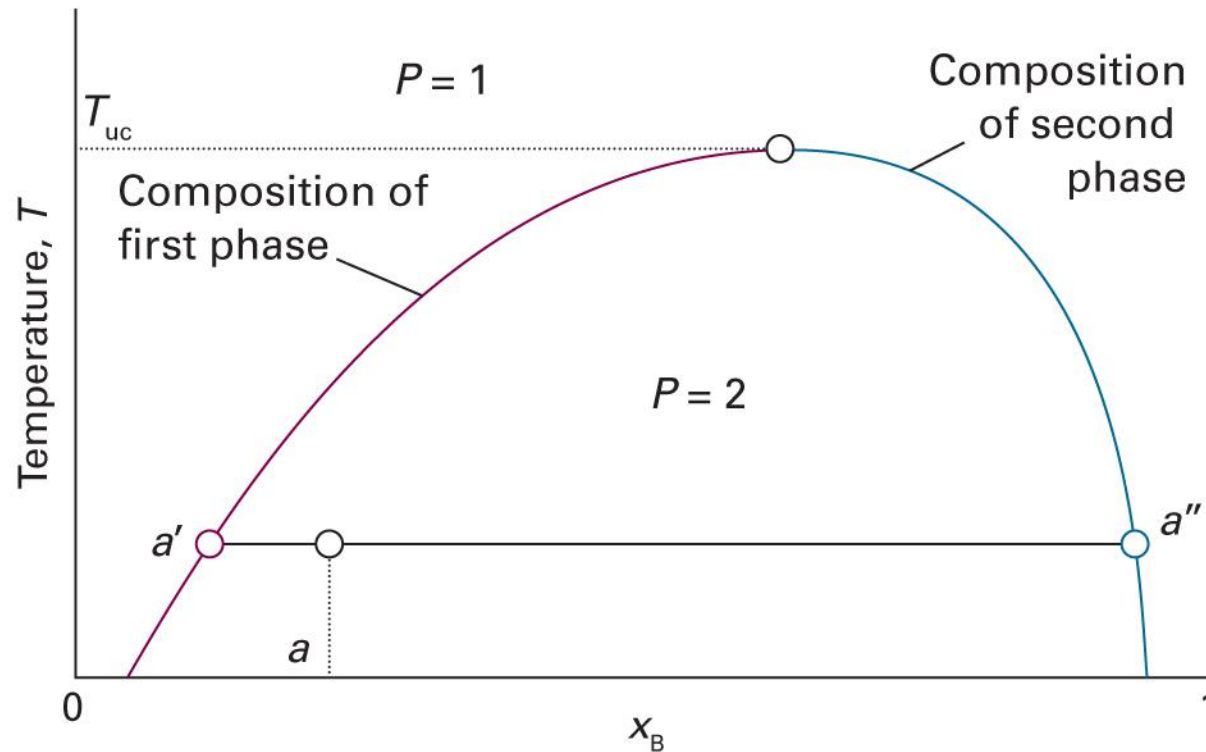


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.6)

Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$

Phase Diagram for Mixture



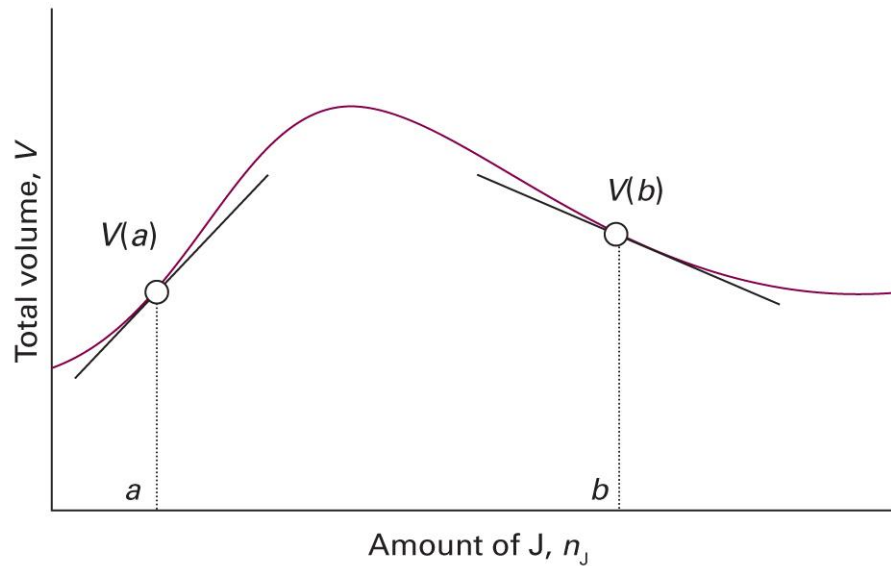
Partial Molar Quantities

$$X_J = \left(\frac{\partial X}{\partial n_J} \right)_{p,T,n'}$$

contribution that a component of a mixture makes
to the total quantity of a sample

Partial Molar Volume

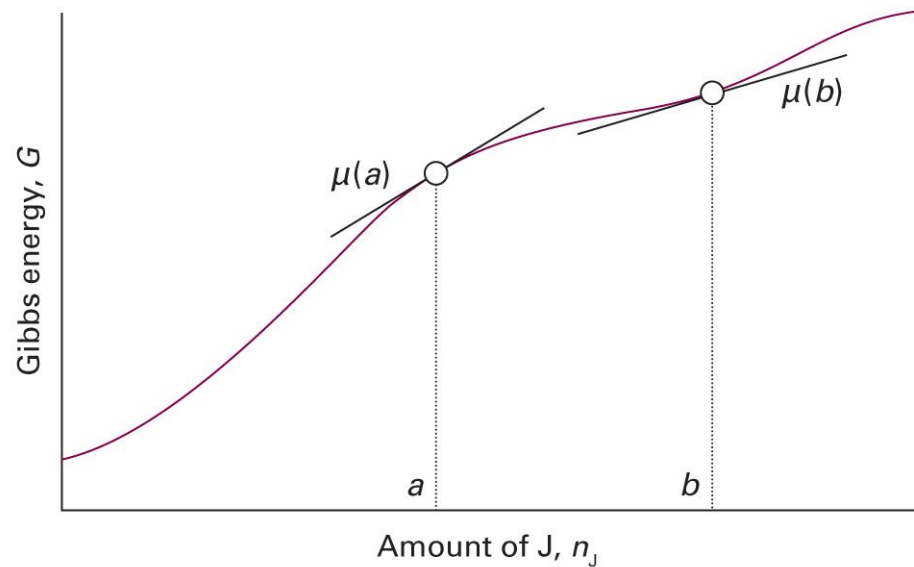
$$V_J = \left(\frac{\partial V}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.2)

Partial Molar Gibbs Energy

$$\mu_J = \left(\frac{\partial G}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.4)

... is Chemical Potential

$$dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

Hence,

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

Gibbs-Duhem Equation

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

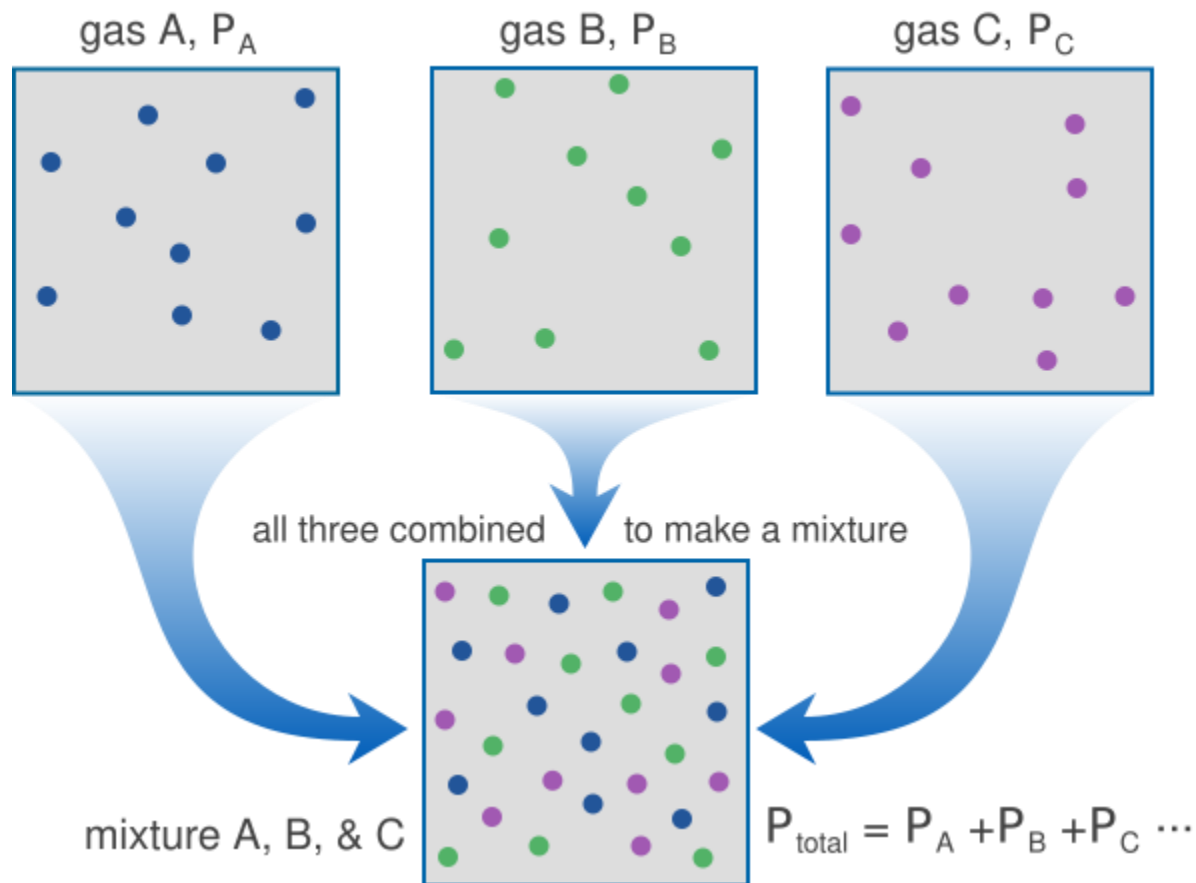
At constant T and p ,

$$\sum_J n_J d\mu_J = 0$$

For a two-component system,

$$n_A d\mu_A + n_B d\mu_B = 0$$

Gas Mixture



Dalton's Law

Partial pressure for species J is

$$p_J = x_J p$$

so

$$p_A + p_B + \cdots = (x_A + x_B + \cdots)p = p$$

Valid for all gases (not only perfect)

Chemical Potential

For a perfect gas,

$$G_m(p) = G_m^\ominus + RT \ln \frac{p}{p^\ominus}$$

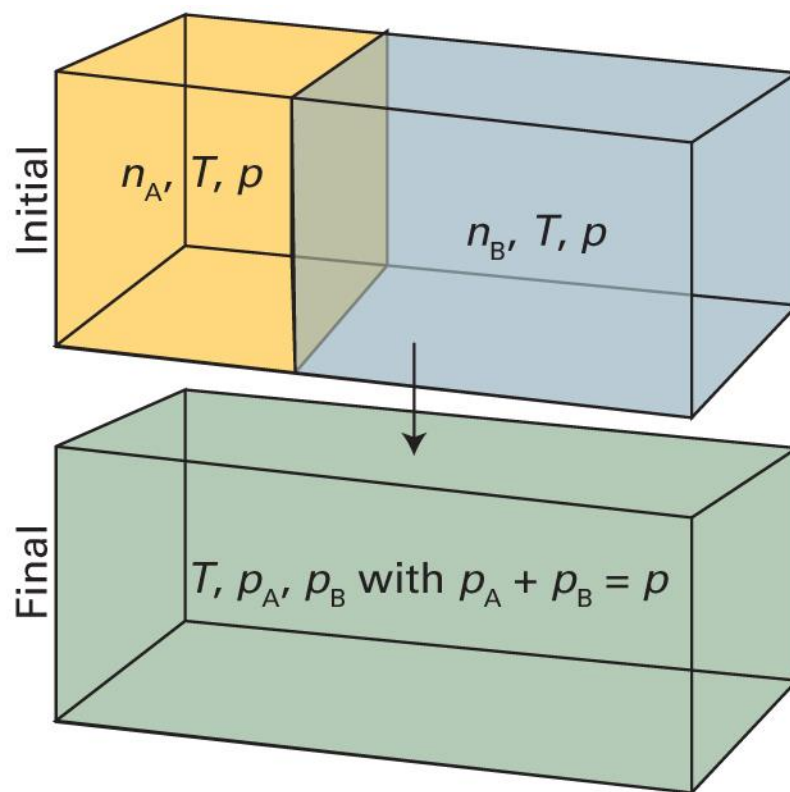
or,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

where $\mu^\ominus$ is the standard chemical potential (at 1 bar).

Mixing of Gases

(at constant T and p)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.6)

Total Gibbs Energy

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

- Initial state

$$G_i = n_A\{\mu_A^\ominus + RT \ln(p/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/p^\ominus)\}$$

- Final state

$$G_f = n_A\{\mu_A^\ominus + RT \ln(p_A/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p_B/p^\ominus)\}$$

Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = G_f - G_i = n_A RT \ln \left(\frac{p_A}{p} \right) + n_B RT \ln \left(\frac{p_B}{p} \right)$$

Using Dalton's law,

$$\Delta_{\text{mix}}G = n_A RT \ln x_A + n_B RT \ln x_B$$

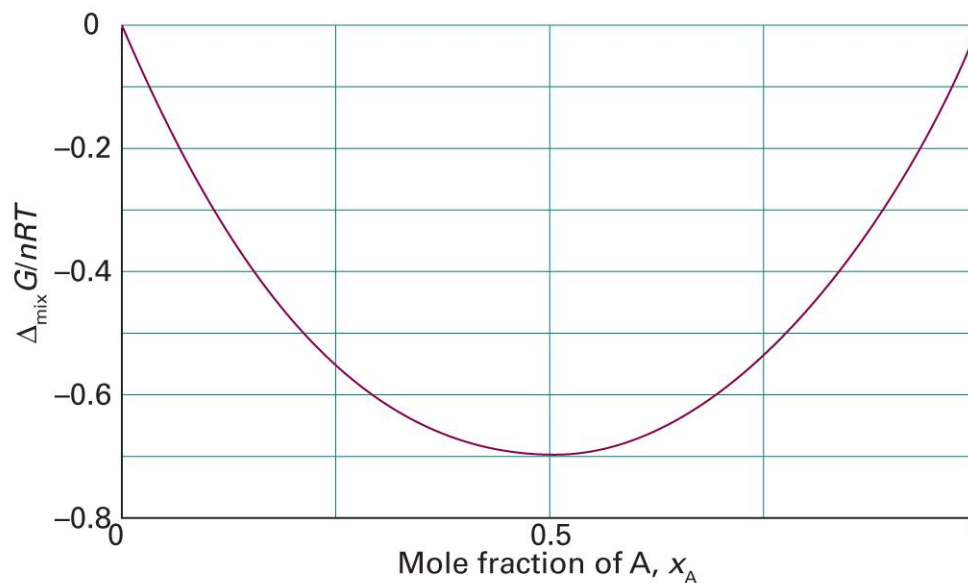
and

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

Note: since $\ln x < 0$ for $x < 1$, $\Delta_{\text{mix}}G < 0$ always.

Gibbs Energy of Mixing

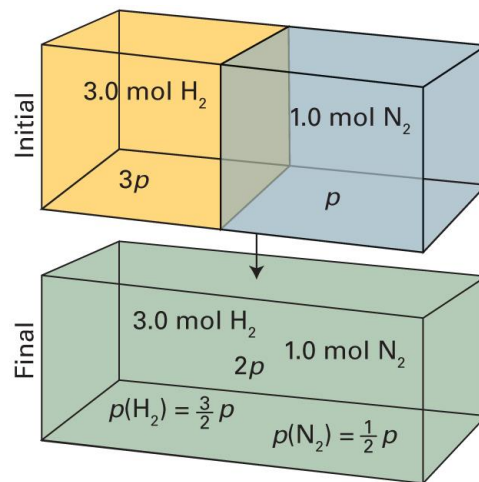
$$\Delta_{\text{mix}}G = nRT\{x_A \ln x_A + (1 - x_A) \ln(1 - x_A)\}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.7)

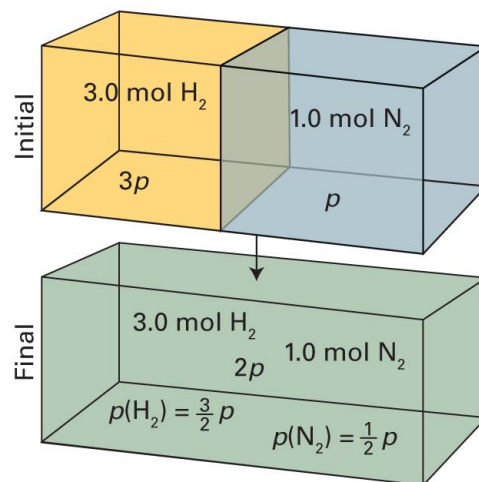
Example 5A.3

A container is divided into two equal compartments. One contains 3.0 mol $\text{H}_2(\text{g})$ at 25°C ; the other contains 1.0 mol $\text{N}_2(\text{g})$ at 25°C . Calculate the Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.7)

Example 5A.3



- H₂: amount 3.0 mol, volume $V \rightarrow 2V$, pressure $3p \rightarrow (3/2)p$
- N₂: amount 1.0 mol, volume $V \rightarrow 2V$, pressure $p \rightarrow (1/2)p$

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

Example 5A.3

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

Total Gibbs energies are

$$G_i = n_A\{\mu_A^\ominus + RT \ln(3p/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/p^\ominus)\}$$

$$G_f = n_A\{\mu_A^\ominus + RT \ln(3p/2p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/2p^\ominus)\}$$

Therefore,

$$\Delta_{\text{mix}}G = G_f - G_i = -n_A RT \ln 2 - n_B RT \ln 2 = -(n_A + n_B) RT \ln 2$$

$$= -(3.00 \text{ mol} + 1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times 0.693$$

$$= -6.9 \text{ kJ}$$



Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

Real Gases

For a perfect gas,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

For a real gas,

$$\mu = \mu^\ominus + RT \ln \frac{f}{p^\ominus}$$

where f is the **fugacity**.

Fugacity and Pressure

For a perfect gas,

$$\mu_2 - \mu_1 = \int_{p_1}^{p_2} V_m dp = RT \int_{p_1}^{p_2} \frac{1}{p} dp = RT \ln \frac{p_2}{p_1}$$

For a real gas, we have the virial expansion:

$$pV_m = RT(1 + B'p + C'p^2 + \dots)$$

Now,

$$\mu_2 - \mu_1 = \int_{p_1}^{p_2} V_m dp = \int_{p_1}^{p_2} \frac{RT}{p} (1 + B'p + C'p^2 + \dots) dp$$

Fugacity and Pressure

$$\begin{aligned}\mu_2 - \mu_1 &= \int_{p_1}^{p_2} \frac{RT}{p} (1 + B'p + C'p^2 + \dots) dp \\&= RT \int_{p_1}^{p_2} \frac{1}{p} dp + B'RT \int_{p_1}^{p_2} 1 dp + C'RT \int_{p_1}^{p_2} p dp + \dots \\&= RT \ln \left(\frac{p_2}{p_1} \right) + B'RT(p_2 - p_1) + \frac{C'RT}{2} (p_2^2 - p_1^2) + \dots \\&\approx RT \ln \left(\frac{p_2}{p_1} \right) + B'RTp_2 + \frac{C'RT}{2} p_2^2 + \dots \text{ (for } p_1 \rightarrow 0\text{)}\end{aligned}$$

Fugacity and Pressure

For $p_1 \rightarrow 0$,

$$\mu_2 - \mu_1 = RT \ln \left(\frac{p_2}{p_1} \right) + B' RT p_2 + \frac{C' RT}{2} p_2^2 + \dots$$

At this limit,

$$\mu_1 \approx \mu_{\text{perfect}} = \mu^\ominus + RT \ln(p_1/p^\ominus)$$

Thus,

$$\mu_2 = \mu^\ominus + RT \ln \left(\frac{p_2}{p^\ominus} \right) + B' RT p_2 + \frac{C' RT}{2} p_2^2 + \dots$$

Fugacity and Pressure

$$\mu = \mu^\ominus + RT \ln \left(\frac{p}{p^\ominus} \right) + B' RT p + \frac{C' RT}{2} p^2 + \dots$$

$$= \mu^\ominus + RT \left\{ \ln \left(\frac{p}{p^\ominus} \right) + B' p + \frac{C'}{2} p^2 + \dots \right\}$$

$$= \mu^\ominus + RT \left\{ \ln \left(\frac{p}{p^\ominus} \right) + \ln(e^{B' p}) + \ln(e^{C' p^2/2}) + \dots \right\}$$

$$= \mu^\ominus + RT \ln \left(\frac{p}{p^\ominus} \underbrace{e^{B' p + C' p^2/2 + \dots}}_{\text{Imperfectness}} \right)$$

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Fugacity and Pressure

$$\mu = \mu^\ominus + RT \ln \left(\frac{p}{p^\ominus} e^{B'p + C'p^2/2 + \dots} \right)$$

$$\mu = \mu^\ominus + RT \ln \frac{f}{p^\ominus}$$

Comparison gives

$$f = p e^{B'p + C'p^2/2 + \dots}$$

Properties of Fugacity

$$f = pe^{B'p+C'p^2/2+\dots}$$

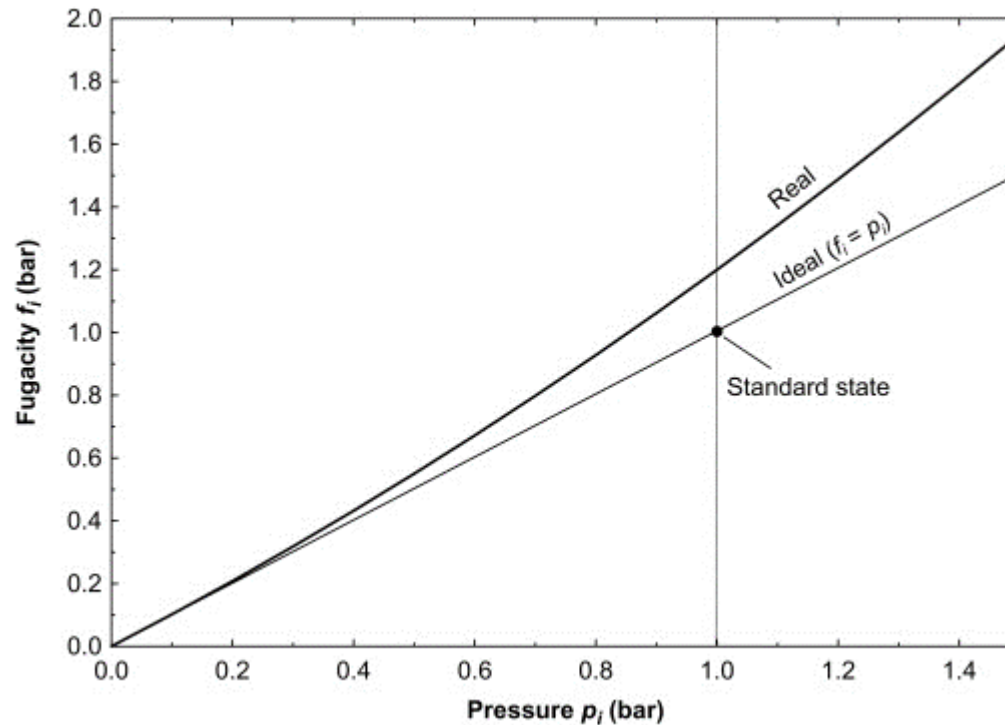
- As $p \rightarrow 0$, $f \rightarrow p$
- For a slightly imperfect gas,

$$f \approx pe^{B'p}$$

- $B' > 0$ (repulsion dominant): $f > p$
- $B' = 0$ (perfect-gas-like): $f = p$
- $B' < 0$ (attraction dominant): $f < p$

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Fugacity and Pressure



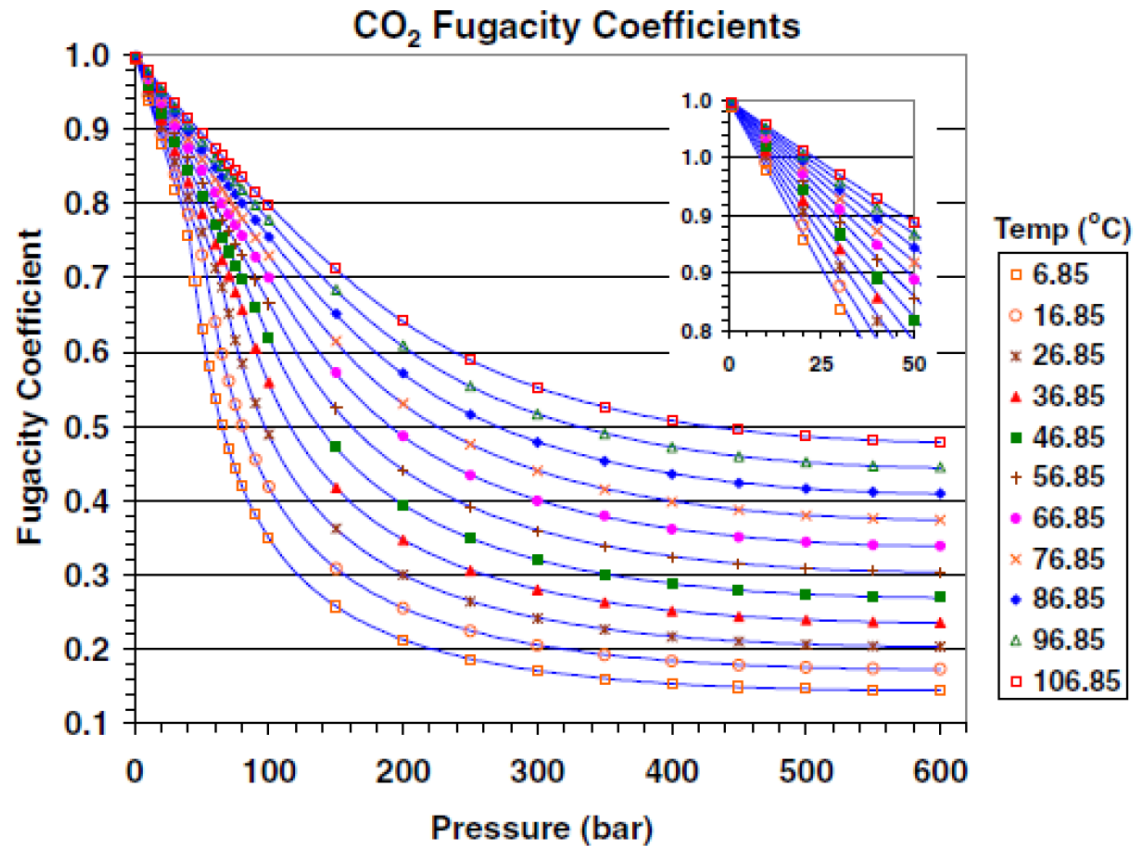
Fugacity Coefficient

$$\phi = \frac{f}{p}$$

- For a perfect gas, $\phi = 1$
- As $p \rightarrow 0$, $\phi \rightarrow 1$
- For a slightly imperfect gas, $\phi \approx e^{B'p}$
 - $B' > 0$ (repulsion-dominant): $\phi > 1$
 - $B' < 0$ (attraction-dominant): $\phi < 1$

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Fugacity Coefficient of CO₂



(Image: <https://dx.doi.org/10.13140/RG.2.2.25544.01285>)

Gibbs Energy: Before Mixing

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln \left\{ \frac{f(p)}{p^\ominus} \right\} = \mu^\ominus + RT \ln \left(\frac{p\phi}{p^\ominus} \right)$$

- Initial state: separated A and B gases at p and T

$$G_i = n_A \left[\mu_A^\ominus + RT \ln \left(\frac{p\phi_A^*}{p^\ominus} \right) \right] + n_B \left[\mu_B^\ominus + RT \ln \left(\frac{p\phi_B^*}{p^\ominus} \right) \right]$$

where ϕ_j^* denotes the fugacity coefficient of species J in its pure form.

Gibbs Energy: After Mixing

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln \left\{ \frac{f(p)}{p^\ominus} \right\} = \mu^\ominus + RT \ln \left(\frac{p\phi}{p^\ominus} \right)$$

- Final state: A and B gases in one container at p and T

From Dalton's law, $p_A = x_A p$ and $p_B = x_B p$

$$G_f = n_A \left[\mu_A^\ominus + RT \ln \left(\frac{x_A p \phi_A}{p^\ominus} \right) \right] + n_B \left[\mu_B^\ominus + RT \ln \left(\frac{x_B p \phi_B}{p^\ominus} \right) \right]$$

Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = G_f - G_i = n_A RT \ln \left(x_A \frac{\phi_A}{\phi_A^*} \right) + n_B RT \ln \left(x_B \frac{\phi_B}{\phi_B^*} \right)$$

$$\Delta_{\text{mix}}G = nRT \left\{ \underbrace{x_A \ln x_A + x_B \ln x_B}_{\text{perfect gas}} + \underbrace{x_A \ln \left(\frac{\phi_A}{\phi_A^*} \right) + x_B \ln \left(\frac{\phi_B}{\phi_B^*} \right)}_{\text{correction}} \right\}$$

In typical conditions, the **correction term** is much smaller and

$$\Delta_{\text{mix}}G = \Delta_{\text{mix}}H - T\Delta_{\text{mix}}S < 0$$

Mixing of Liquids

When A is a pure substance,

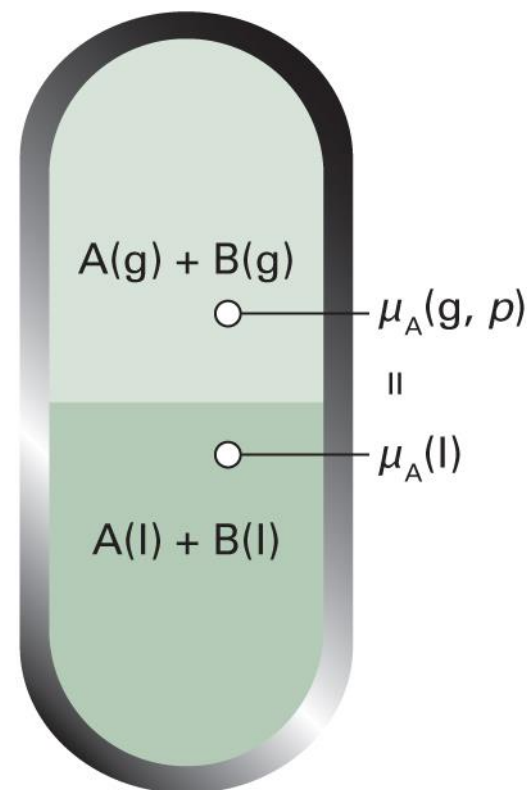
$$\underbrace{\mu_A^*(l)}_{\text{liquid}} = \underbrace{\mu_A^\ominus(g) + RT \ln(p_A^*/p^\ominus)}_{\text{(perfect) gas}}$$

If A is mixed with another substance B,

$$\mu_A(l) = \mu_A^\ominus(g) + RT \ln(p_A/p^\ominus)$$

Subtraction leads to

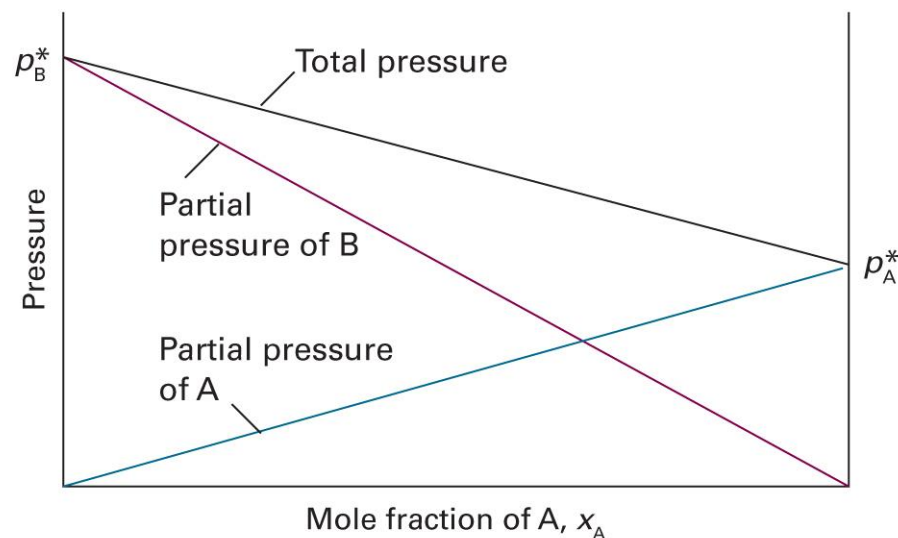
$$\mu_A(l) = \mu_A^*(l) + RT \ln(p_A/p_A^*)$$



Raoult's Law: Ideal Solution

$$p_A = x_A p_A^*$$

- **Ideal solutions:** mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.11)

Chemical Potential

$$\mu_A(l) = \mu_A^*(l) + RT \ln(p_A/p_A^*)$$

For an ideal solution,

$$p_A = x_A p_A^*$$

Hence,

$$\mu_A(l) = \mu_A^*(l) + RT \ln x_A$$

As $x_A < 1$, $\mu_A(l) < \mu_A^*(l)$ and mixed phase is more stable

Ideal Solutions

- Before mixing:

$$G_i = n_A \mu_A^* + n_B \mu_B^*$$

- After mixing:

$$G_f = n_A \mu_A + n_B \mu_B$$

$$= n_A (\mu_A^* + RT \ln x_A) + n_B (\mu_B^* + RT \ln x_B)$$

- Gibbs energy of mixing

$$\Delta_{\text{mix}} G = G_f - G_i = n_A RT \ln x_A + n_B RT \ln x_B$$

$$\Delta_{\text{mix}} G = nRT(x_A \ln x_A + x_B \ln x_B)$$

Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

(Same with perfect gas mixtures)

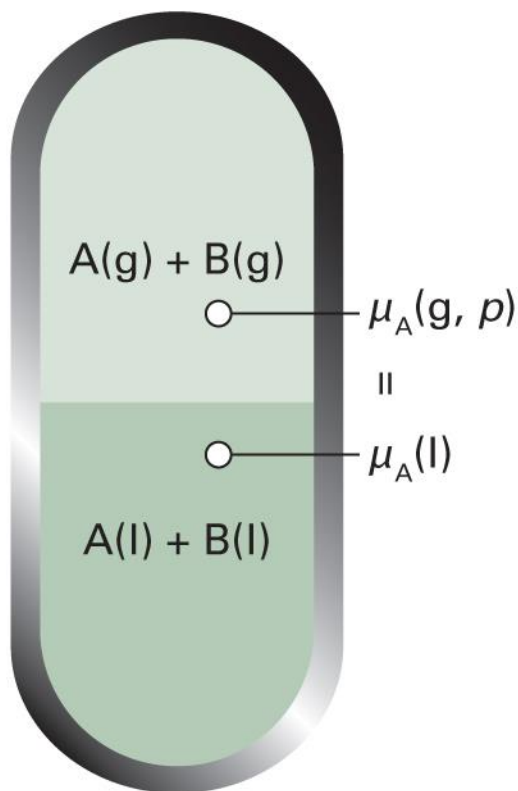
Brief Illustration 5B.1

Consider a mixture of benzene and methylbenzene, which form an approximately ideal solution, and suppose 1.0 mol $\text{C}_6\text{H}_6(l)$ is mixed with 2.0 mol $\text{C}_6\text{H}_5\text{CH}_3(l)$. For the mixture, $x_{\text{benzene}} = 0.33$ and $x_{\text{methylbenzene}} = 0.67$. What are the molar Gibbs energy and entropy of mixing at 25°C?

$$\begin{aligned}\Delta_{\text{mix}}G/n &= (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times (0.33 \ln 0.33 + 0.67 \ln 0.67) \\ &= -1.6 \text{ kJ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta_{\text{mix}}S/n &= -(8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (0.33 \ln 0.33 + 0.67 \ln 0.67) \\ &= 5.3 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

Liquid-Vapor Equilibrium



- Constraints

$$\mu_A(g) = \mu_A(l)$$

$$\mu_B(g) = \mu_B(l)$$

- May have $x_A(g) \neq x_A(l)$

Brief Illustration 5A.3

Consider a mixture of benzene and methylbenzene, which form an approximately ideal solution. The vapour pressure of pure benzene at 20°C is 75.0 Torr and that of pure methylbenzene is 21.0 Torr at the same temperature. What are the mole fractions in the vapour when the two liquids are in an equimolar mixture?

In an equimolar mixture,

$$x_{\text{benzene}}(l) = x_{\text{methylbenzene}}(l) = 0.5$$

Thus,

$$p_{\text{benzene}} = 75.0 \text{ Torr} \times 0.5 = 37.5 \text{ Torr}$$

$$p_{\text{methylbenzene}} = 21.0 \text{ Torr} \times 0.5 = 10.5 \text{ Torr}$$

Brief Illustration 5A.3

The total pressure is $37.5 \text{ Torr} + 10.5 \text{ Torr} = 48.0 \text{ Torr}$.

From Dalton's law,

$$x_{\text{benzene}}(g) = (37.5 \text{ Torr}) / (48.0 \text{ Torr}) = 0.78$$

$$x_{\text{methylbenzene}}(g) = (10.5 \text{ Torr}) / (48.0 \text{ Torr}) = 0.22$$

Vapor Pressures

From Raoult's law,

$$p_A = x_A p_A^*, \quad p_B = x_B p_B^*$$

The total vapor pressure is

$$p = p_A + p_B = x_A p_A^* + x_B p_B^* = x_A p_A^* + (1 - x_A) p_B^*$$

which becomes

$$p = p_B^* + (p_A^* - p_B^*) x_A$$

Vapor Compositions

$$p = p_B^* + (p_A^* - p_B^*)x_A$$

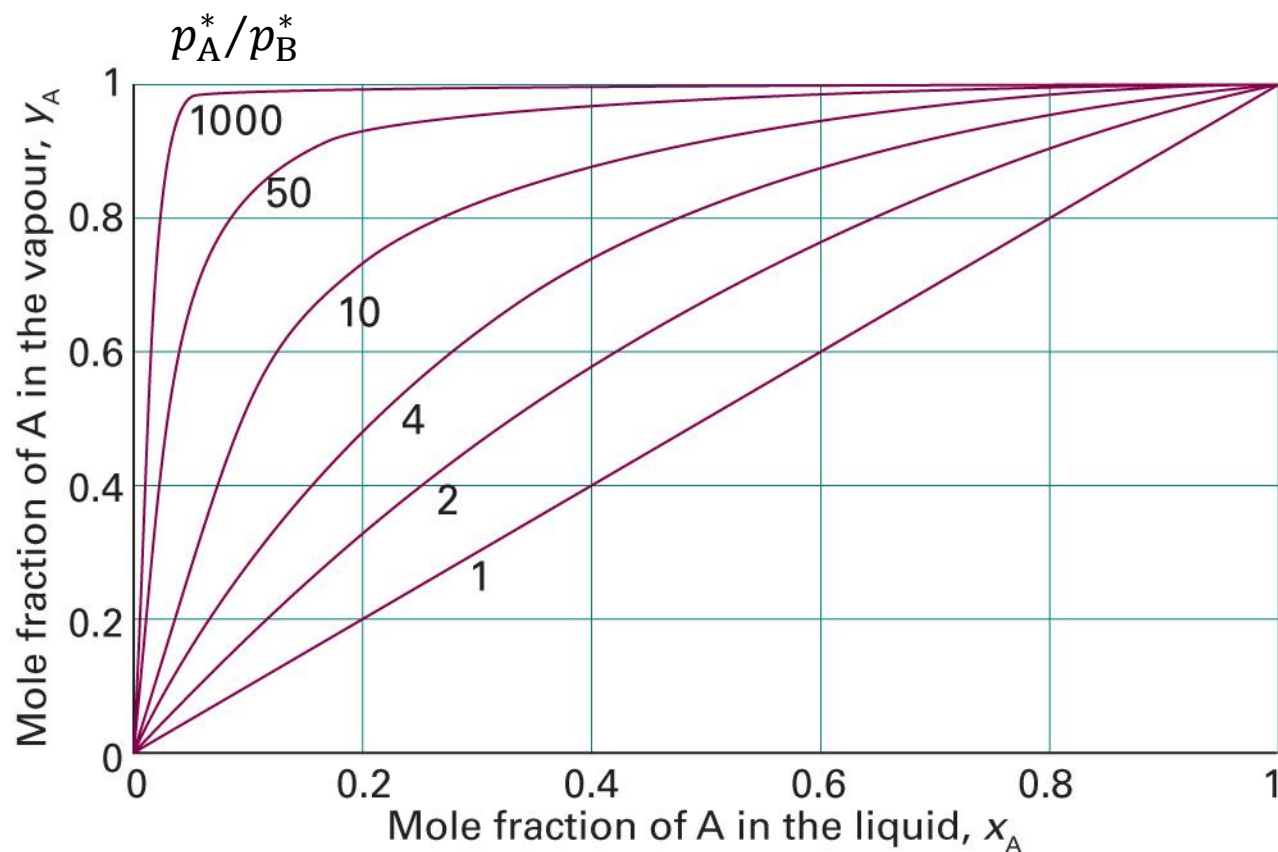
Denote the mole fractions in the vapor as y_A and y_B :

$$y_A = p_A/p, \quad y_B = p_B/p$$

Now,

$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*)x_A}, \quad y_B = 1 - y_A$$

Vapor Compositions



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.2)

x_A as a Function of y_A

$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*)x_A}$$

$$p_B^* y_A + (p_A^* - p_B^*) x_A y_A = x_A p_A^*$$

$$p_B^* y_A = x_A \{p_A^* - (p_A^* - p_B^*) y_A\}$$

Hence,

$$x_A = \frac{y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

Total Vapor Pressure

$$x_A = \frac{y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

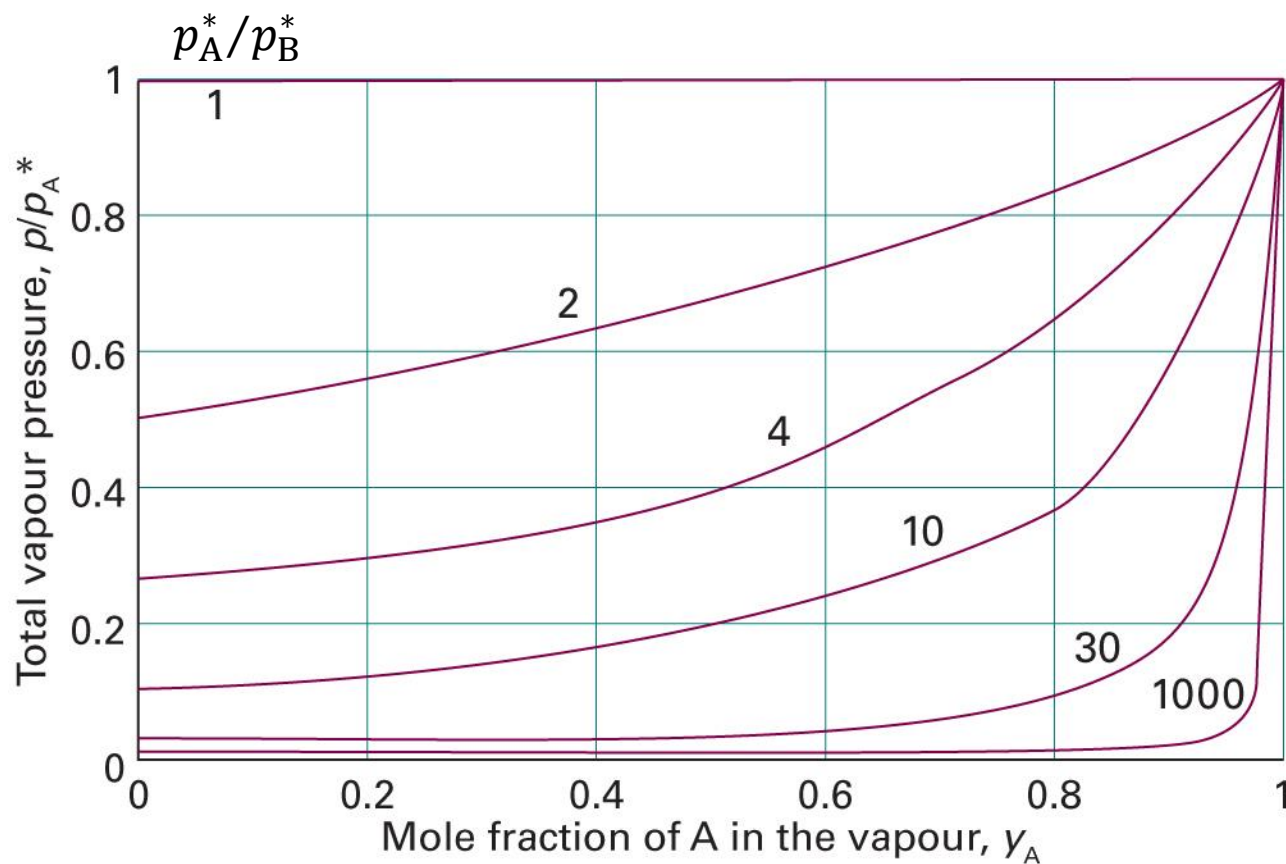
The total vapor pressure is

$$p = p_B^* + (p_A^* - p_B^*) x_A = p_B^* + \frac{(p_A^* - p_B^*) y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

which leads to

$$p = \frac{p_A^* p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

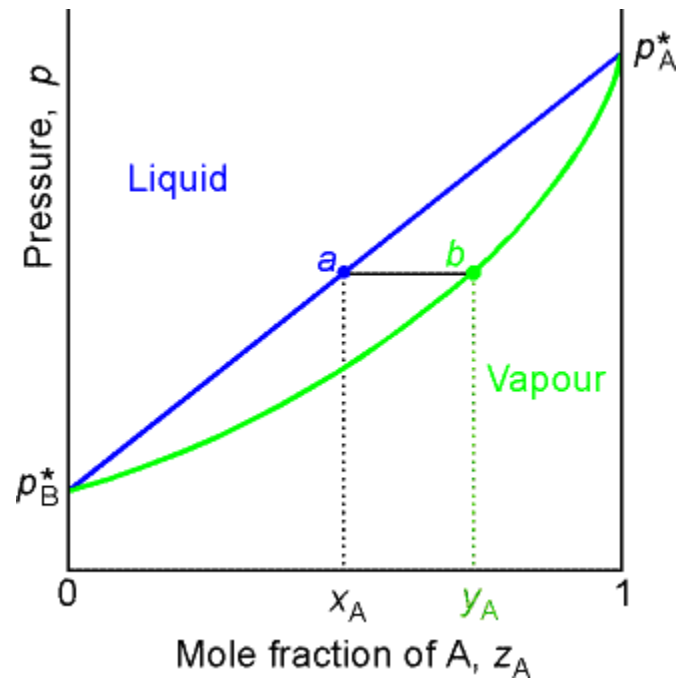
Total Vapor Pressure



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5C.3)

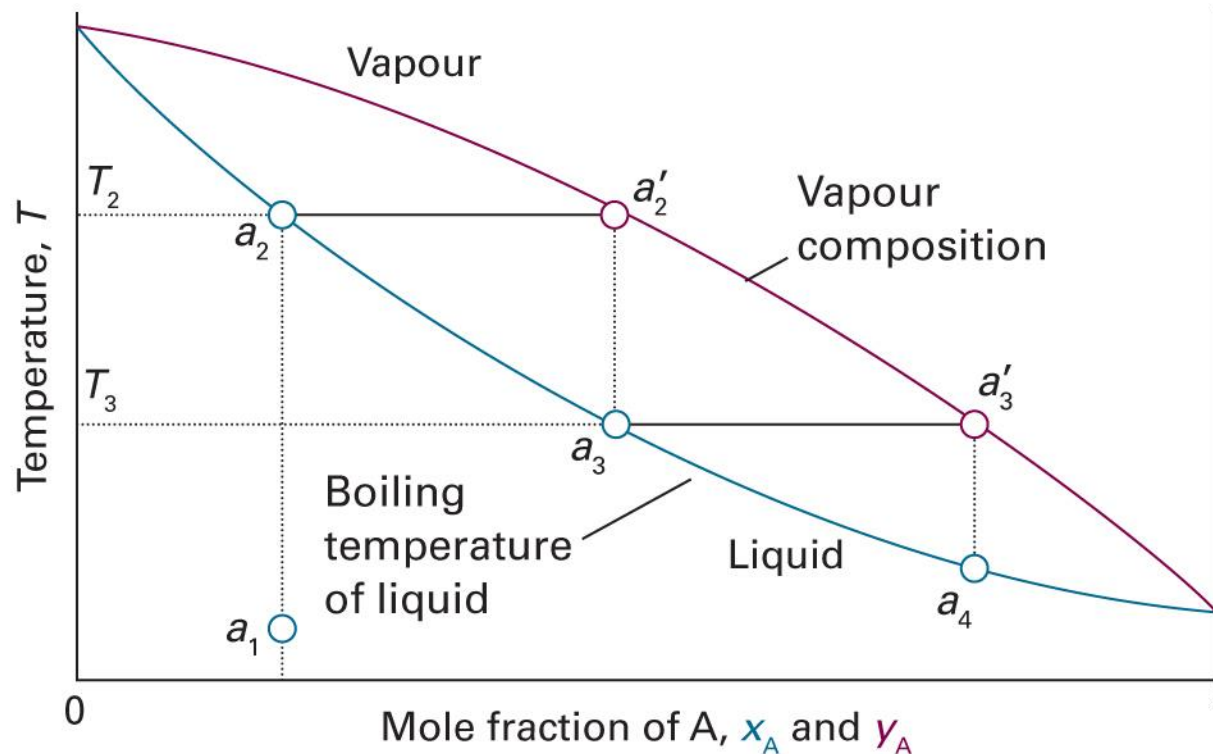
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Pressure-Composition Diagram



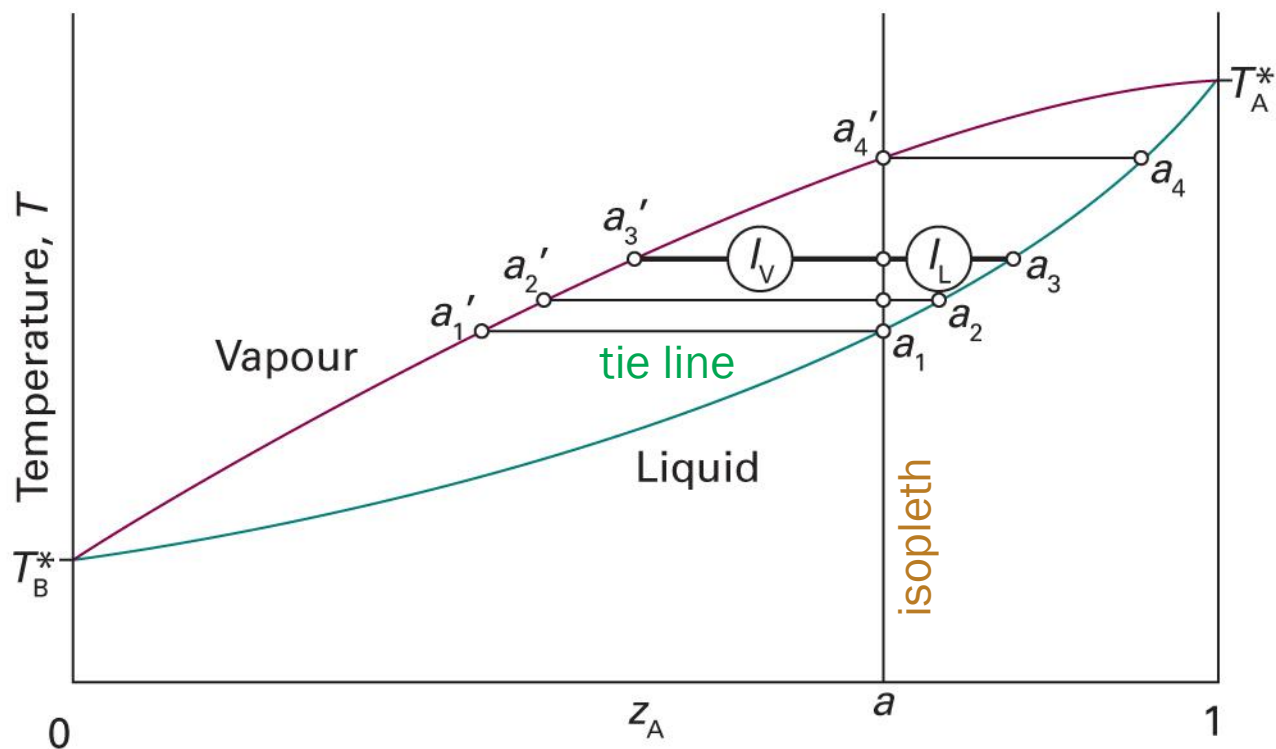
Temperature-Composition Diagram

$$p_A^* > p_B^* \text{ (A is more volatile)}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.4)

Interpretation



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.7)

Lever Rule

- Total amount of A molecules: $n_A = n_{A,l} + n_{A,g}$
- Total amount of molecules in liquid: $n_l = n_{A,l} + n_{B,l}$
- Total amount of molecules in gas: $n_g = n_{A,g} + n_{B,g}$

$$n_{A,l} = x_A n_l, \quad n_{A,g} = y_A n_g$$

Hence,

$$n_A = x_A n_l + y_A n_g$$

Lever Rule

$$n_A = x_A n_l + y_A n_g$$

On the other hand,

$$n_A = n z_A = z_A n_l + z_A n_g$$

Equating the two equations,

$$n_l(z_A - x_A) = n_g(y_A - z_A)$$

$$n_l l_l = n_g l_g$$

